



Deep Learning

Lecture 1.1: Introduction and Class Logistics





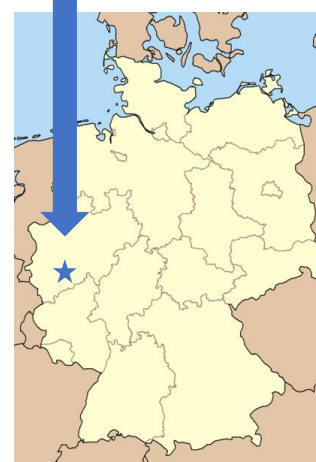
Who am I: Fuxin Li

**Oregon State
University
2015-now**



**Georgia Institute
of Technology
Postdoc 2011-2012
Research Faculty 2012-2015**

**University of Bonn
Postdoc 2008-2010**



**Chinese Academy of Sciences
Ph.D. 2009**



**Zhejiang University
BS 2001**

Research Area: Machine Learning -> Computer Vision -> Robotic Perception

Today's Learning Objectives

Be able to answer:

- What is deep learning?
- Why and when should we care about it?
- What are some problems with deep learning?

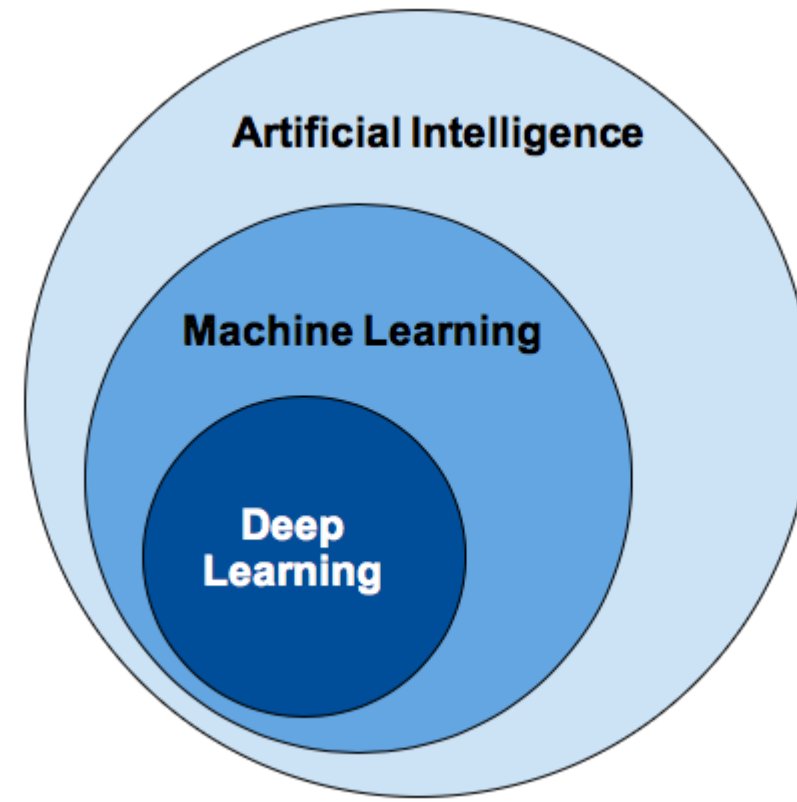
- How is this class structured?
- How will I be assessed / graded?





What is deep learning?

Deep Learning - A set of interrelated model classes composed of hierarchies of differentiable, parameterized, non-linear functions.





What is deep learning?

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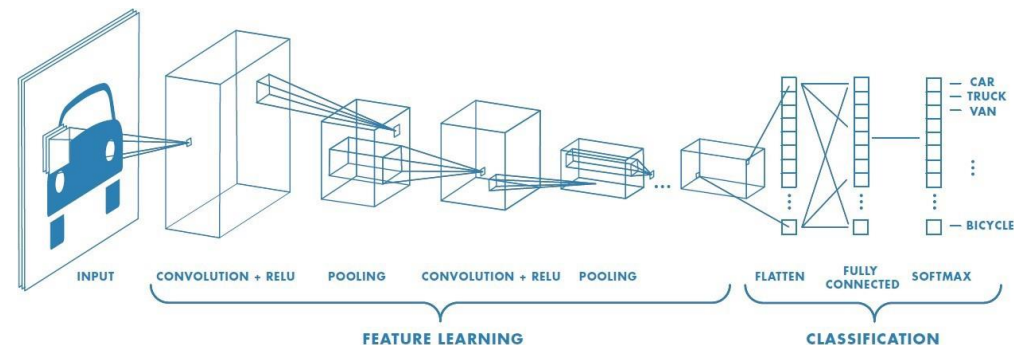
Rapid advance due to -

- Availability of large datasets
- GPU-enabled training (matrix multiply)
- Proliferation of excellent libraries
- Modularity of network components
- Strong performance (when bias is right)

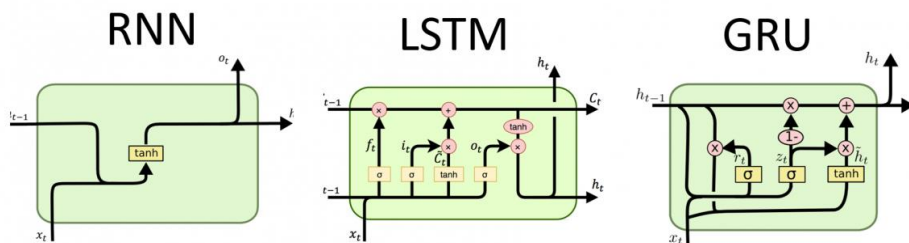


What is deep learning? Example model types.

Convolutional Neural Network



Recurrent Neural Network



Transformer Networks

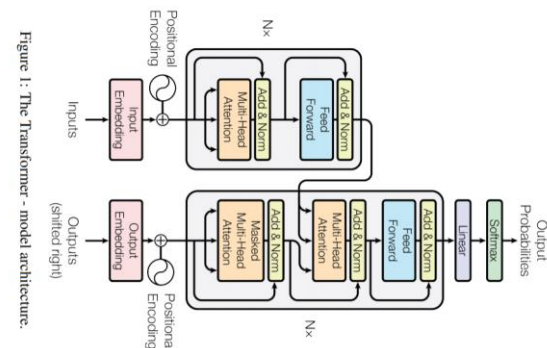


Figure 1: The Transformer - model architecture.



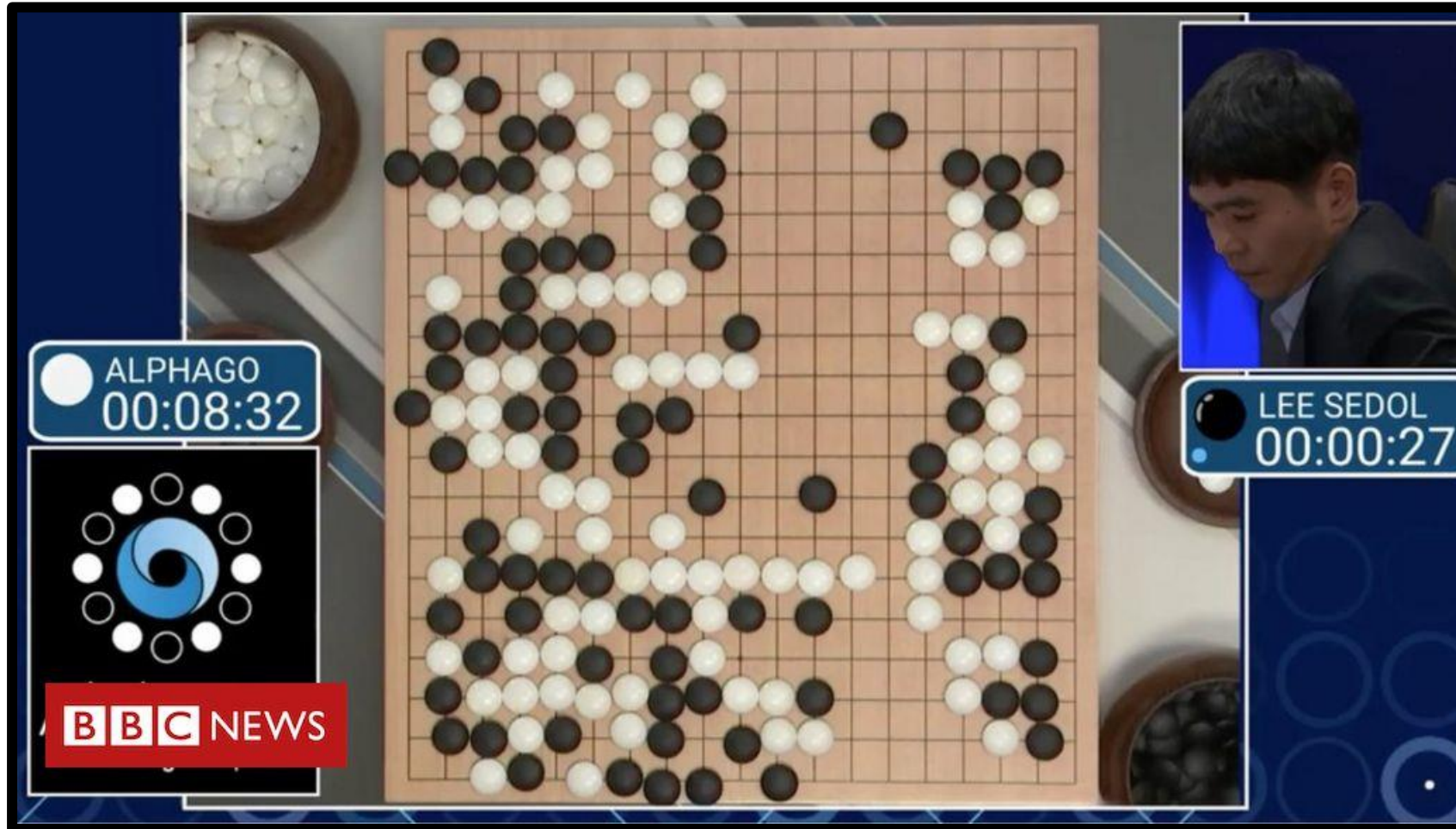
What is deep learning? Recent public successes.



Been neural since 2016.

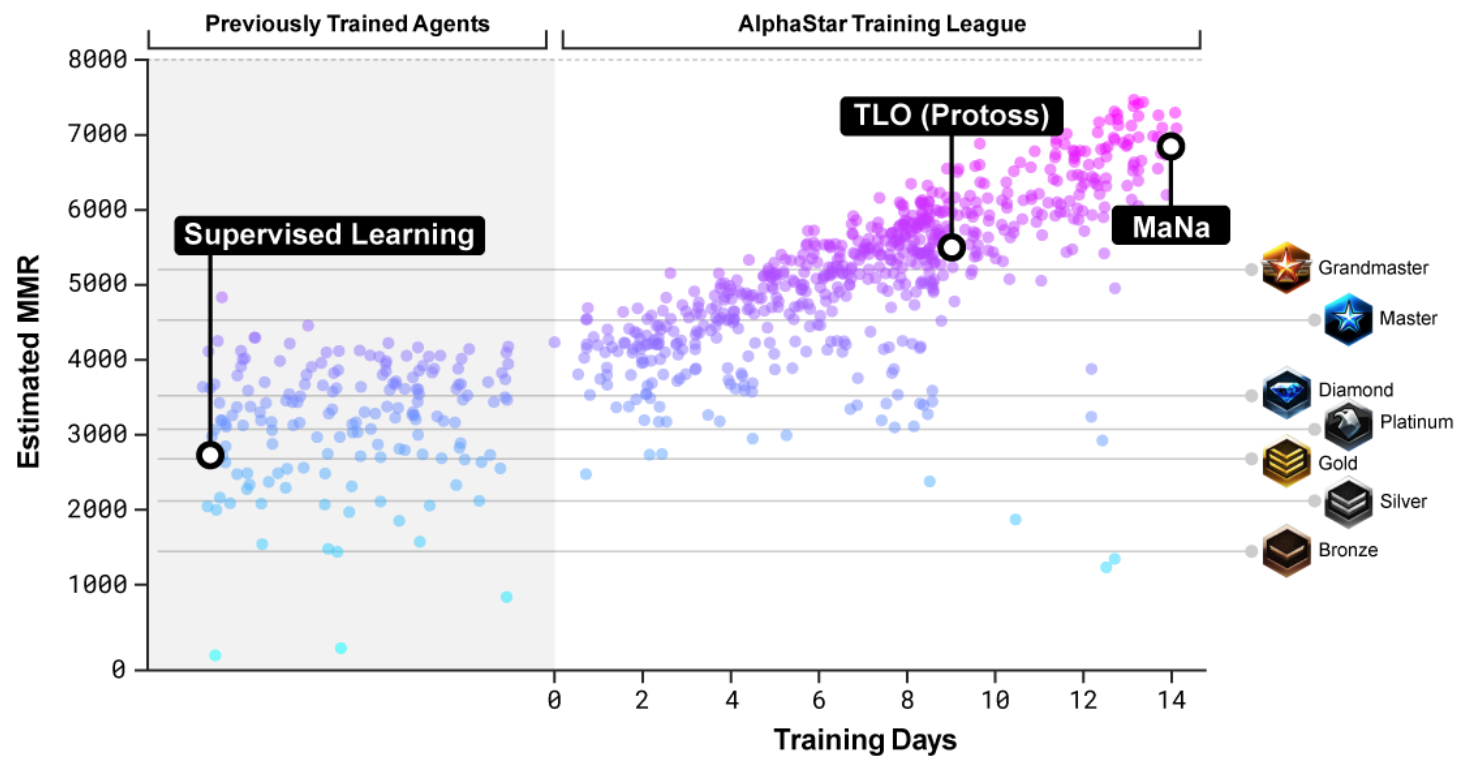


What is deep learning? Recent public successes.





What is deep learning? Recent public successes.





What is deep learning? Recent public successes.





What is deep learning? Recent public successes.





What is deep learning? Recent public successes.

TEXT PROMPT

an illustration of a baby daikon radish in a tutu walking a dog

AI-GENERATED
IMAGES



TEXT PROMPT

a store front that has the word 'openai' written on it. . . .

AI-GENERATED
IMAGES



What is deep learning? Not perfect.

Ligature



Jellyfish (99%)



Painting

Goldfish (99%)



Highway

Dam (99%)



Garlic Bread

Hotdog (99%)

Dragonfly



Manhole Cover



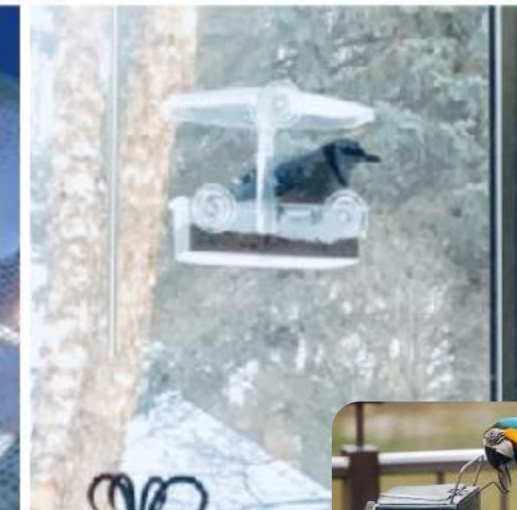
Bullfrog

Fox Squirrel



Monarch Butterfly

Washing Machine



Jay

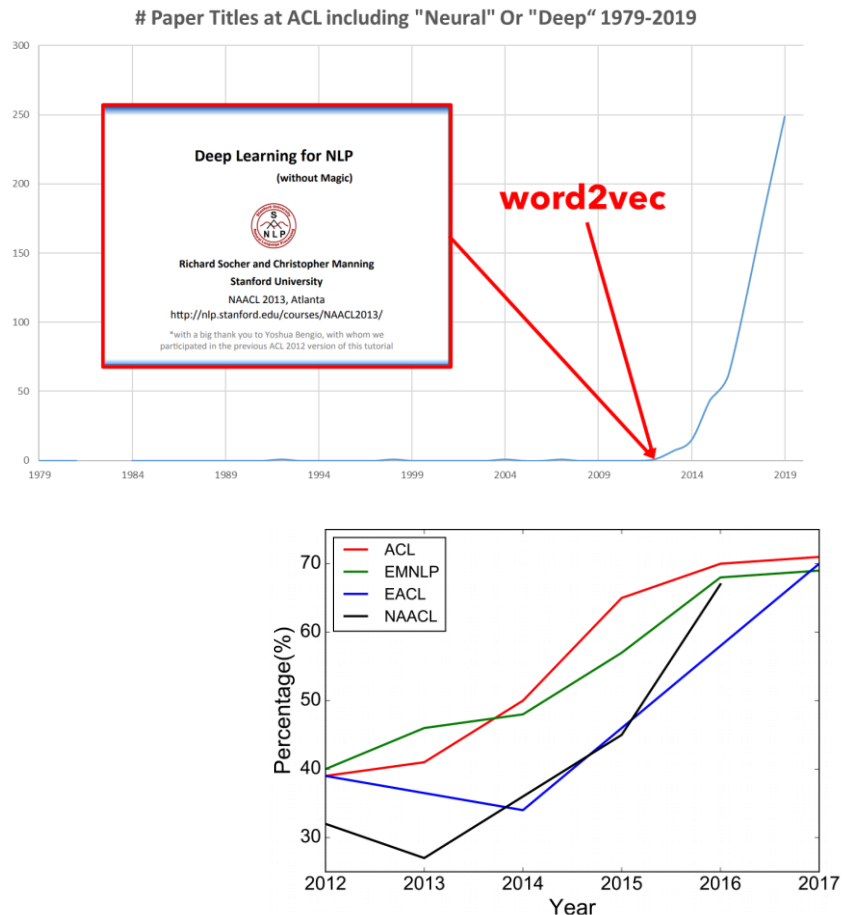
Jeep





What is deep learning? Academic attention.

Growth of deep learning in NLP



Impact of deep learning in Computer Vision

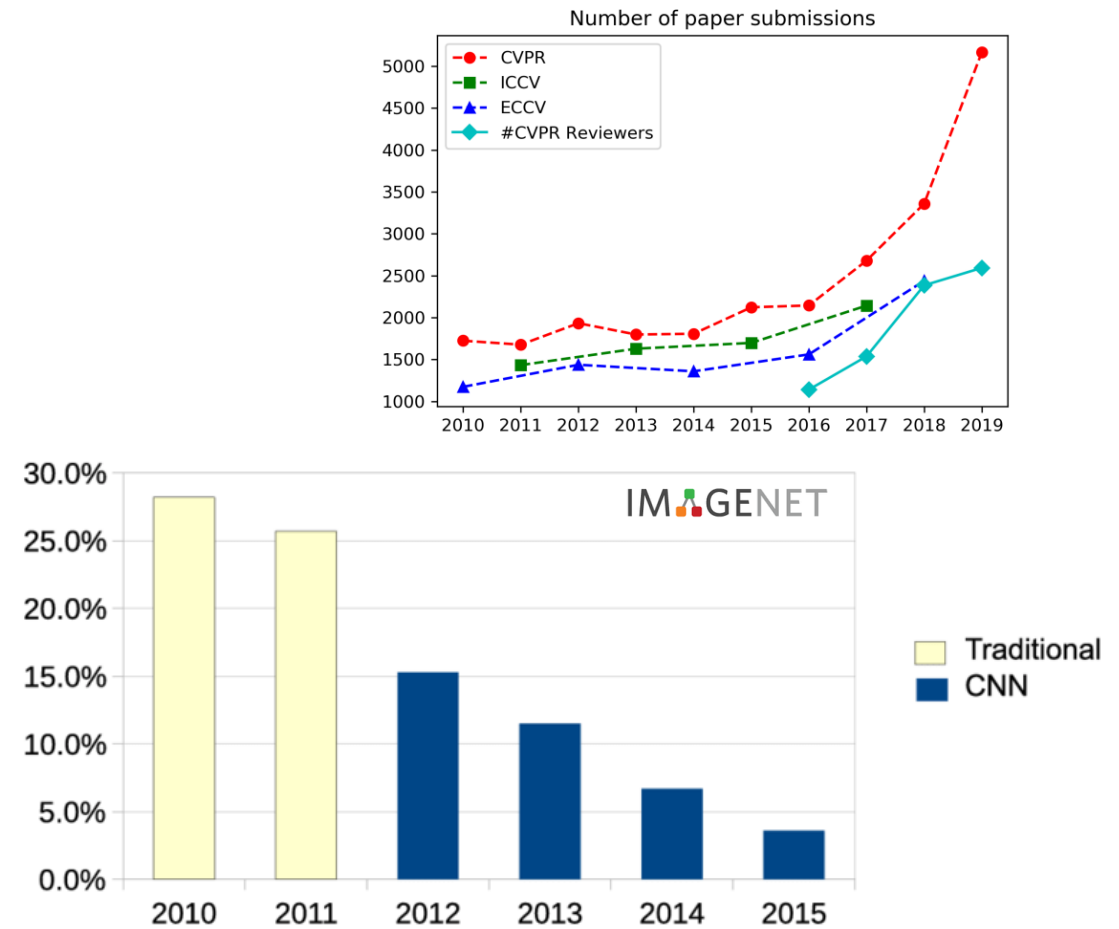


Fig. 1: Percentage of deep learning papers in ACL, EMNLP, EACL, NAACL over the last 6 years (long papers).



What is deep learning? Public attention.

Interest over time ?





What is deep learning?

Deep Learning - A set of interrelated model classes composed of hierarchies of differentiable, parameterized, non-linear functions.

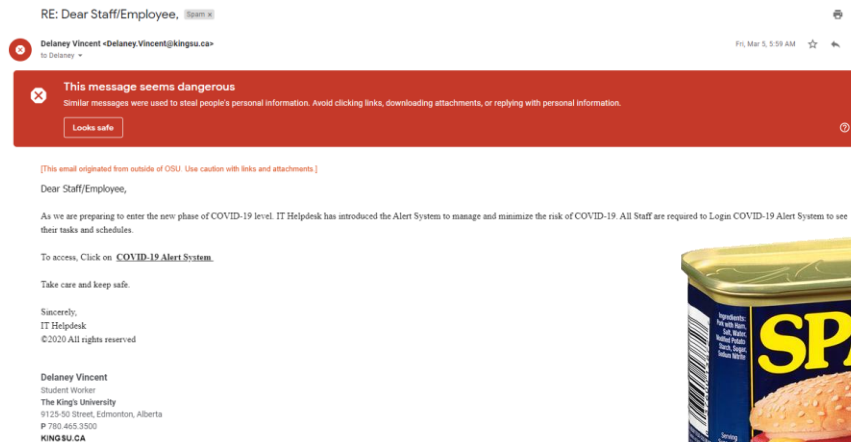
Why isn't this definition more concrete?

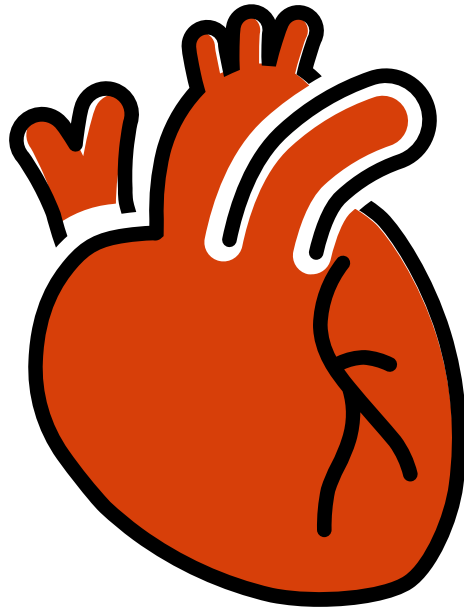
If this is "deep" learning, what is "shallow" learning?

Why hierarchical compositions of functions?

Why must they be differentiable?

- **Thousands of machine learning algorithms**
 - Hundreds new every year
- **Decades of ML research oversimplified:**
 - All of Machine Learning:
 - Learn a mapping from input to output **$g: X \rightarrow Y$**
X: emails, Y: {spam, notspam}



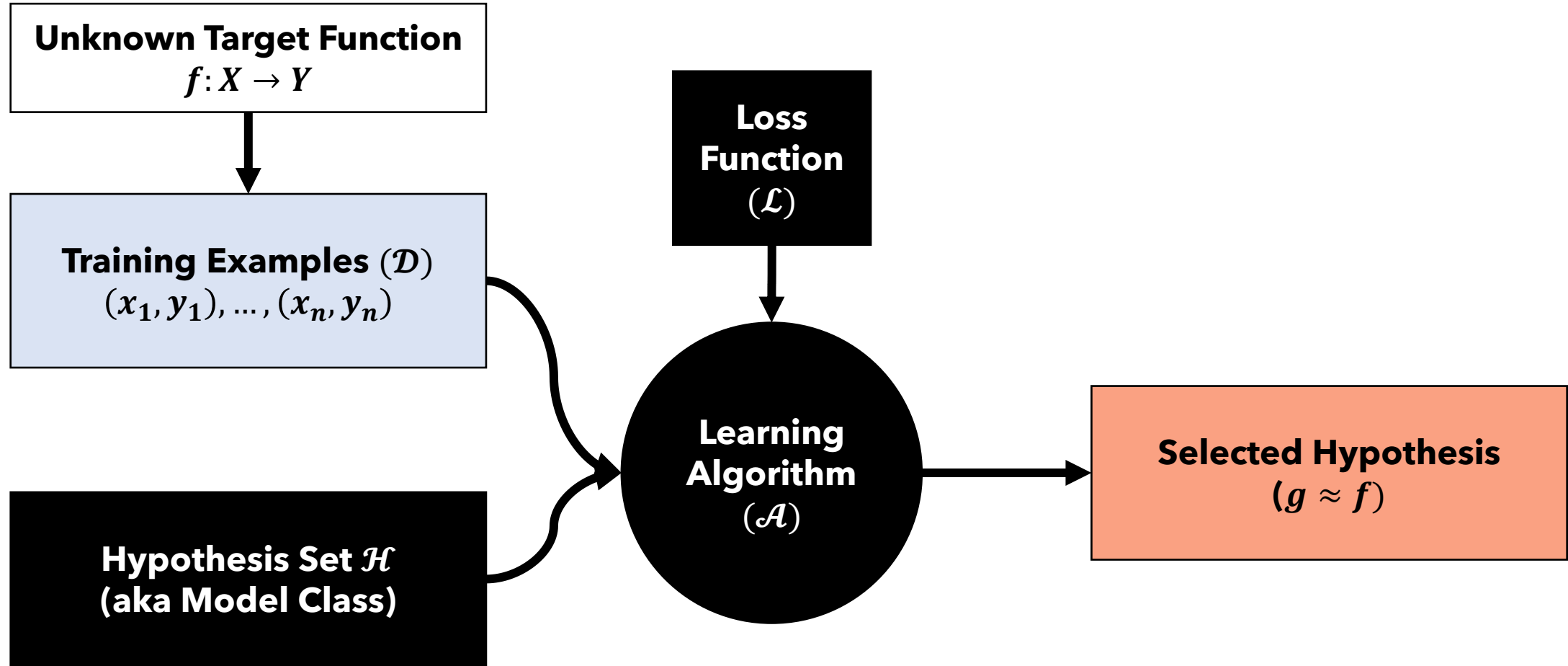


Function Approximation

Find function $\mathbf{g}: X \rightarrow Y$ from hypothesis space $\mathcal{H}$ that best approximates $\mathbf{f}: X \rightarrow Y$

**Usually unknown but
have samples from it**





Observations of some phenomenon



Designed by us

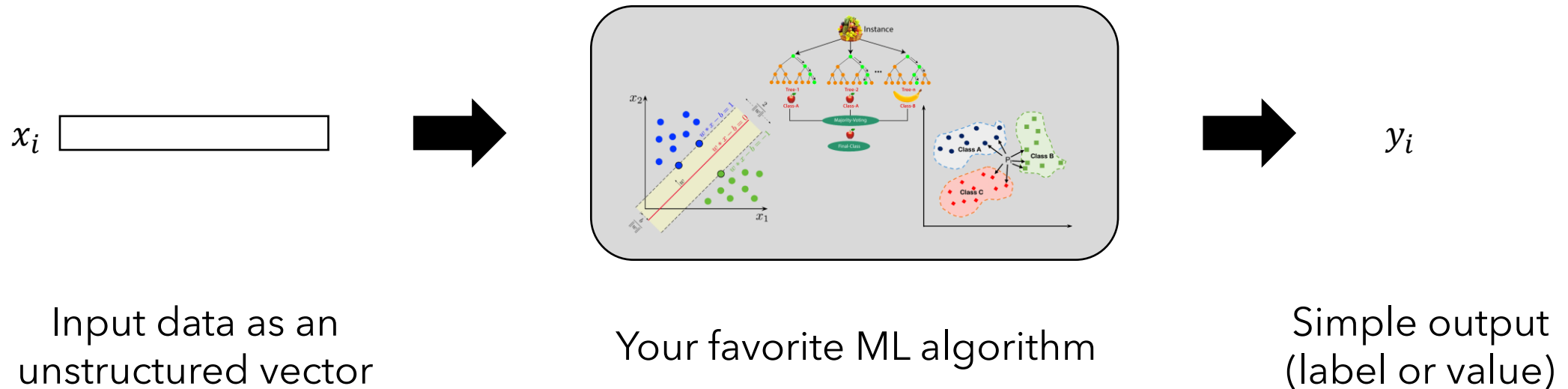


Result of our learning



What is deep learning? Shallow vs Deep.

“Deep” learning implies existence of “shallow” learning:



If your data looks like this, you don't need deep learning!

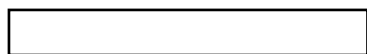


What is deep learning? Shallow vs Deep.

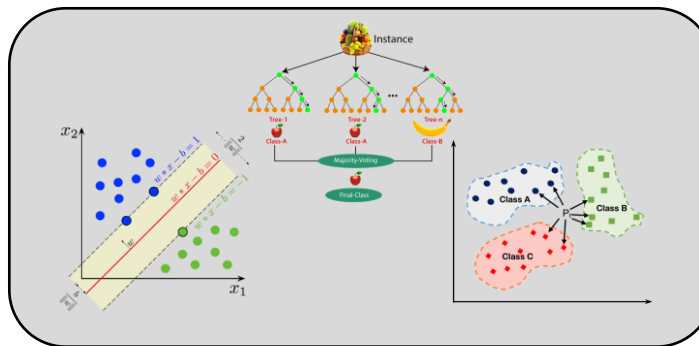
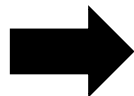
How to fit complex structured problems into this framework?



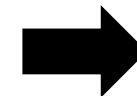
x_i



Input data as an
unstructured vector



Your favorite ML algorithm



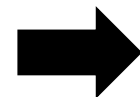
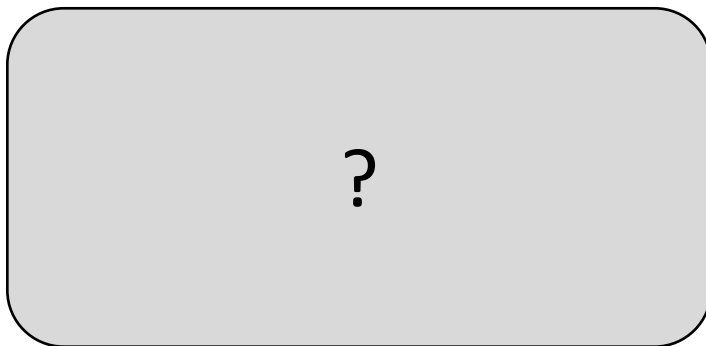
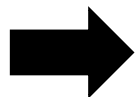
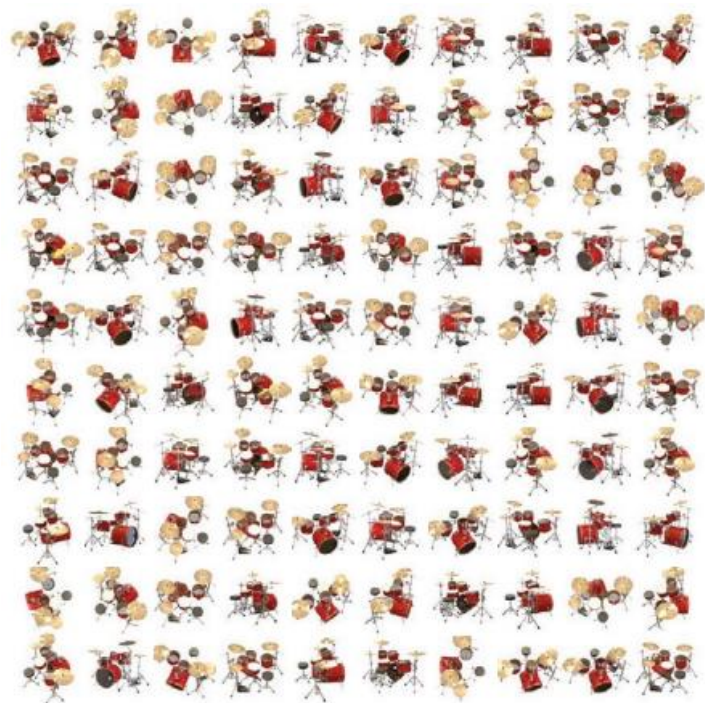
y_i

Simple output
(label or value)



What is deep learning? Shallow vs Deep.

How to fit complex structured problems into this framework?

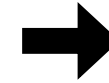
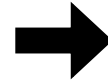
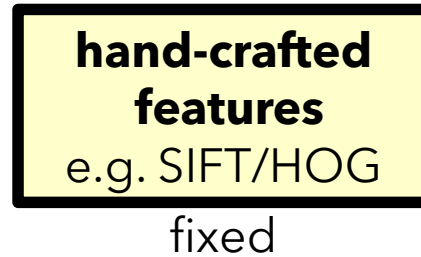
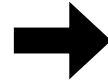




What is deep learning? Shallow vs Deep.

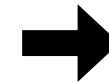
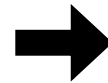
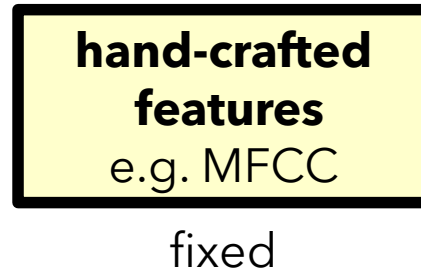
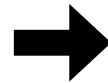
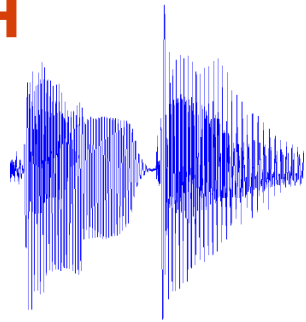
Traditional "Shallow" Learning for Structured Domains

VISION



"car"

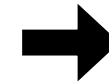
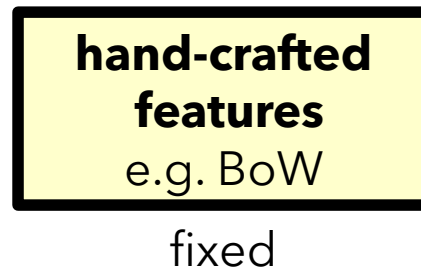
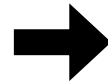
SPEECH



\ 'd ē p \

LANGUAGE

This burrito place
is yummy and fun!

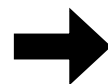


"+"



What is deep learning? Shallow vs Deep.

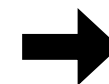
VISION



**hand-crafted
features**
e.g. SIFT/HOG
fixed



**your favorite
classifier**
learned



"car"

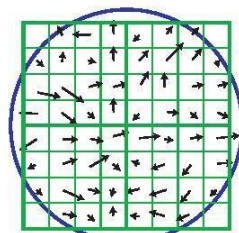
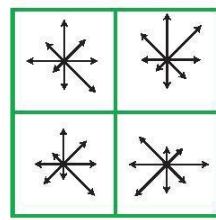
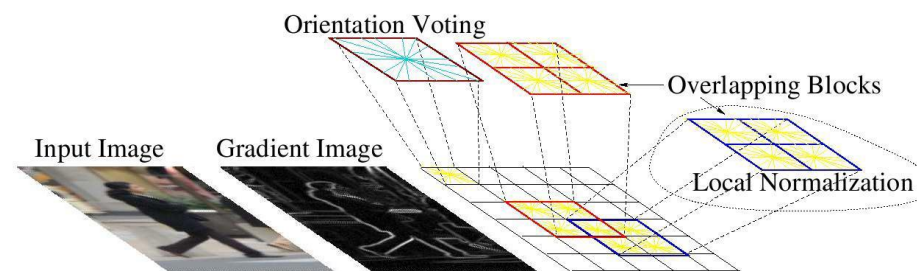
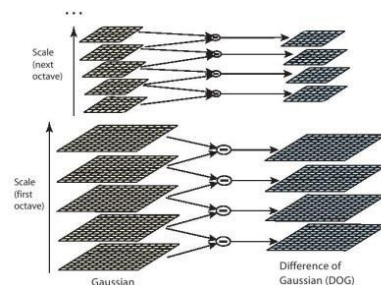


Image gradients

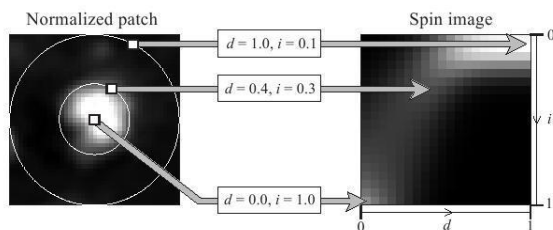


Keypoint descriptor

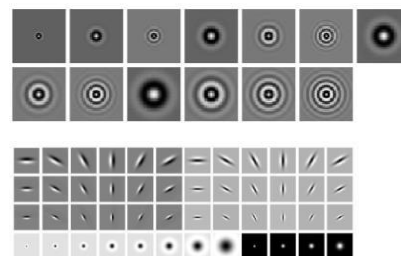
SIFT



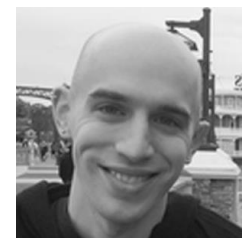
HoG



Spin Images



Textons



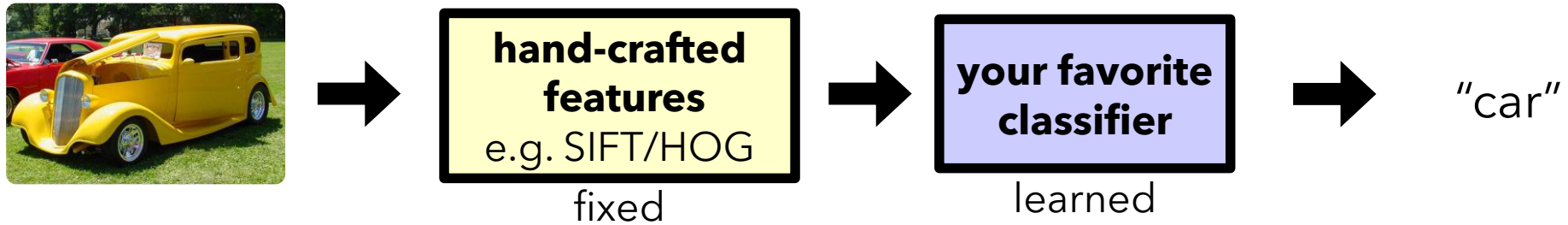
Local Binary Patterns

● ● ●
Plenty more

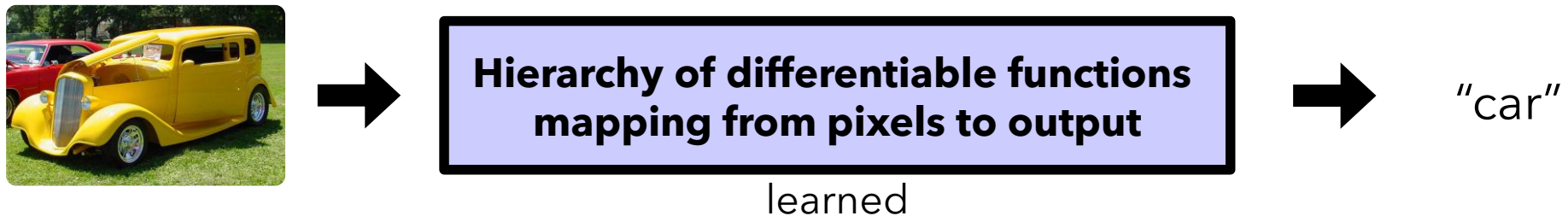


What is deep learning? Shallow vs Deep.

VISION



Deep VISION

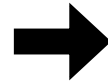


"Because gradient descent is better than you"
-- Yann LeCun

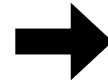


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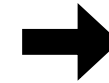
VISION



**hand-crafted
features**
e.g. SIFT/HOG
fixed

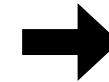
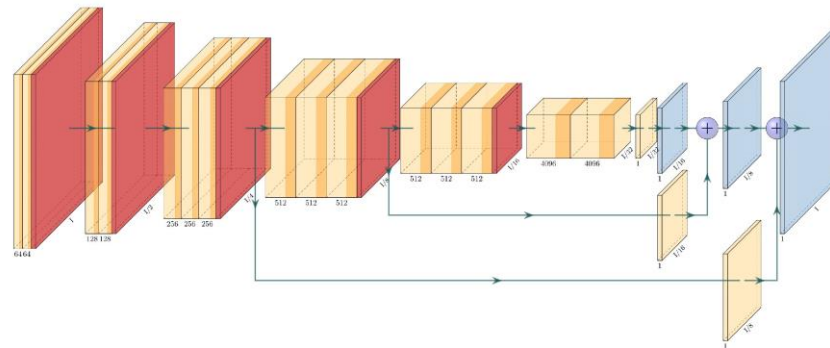
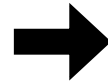


**your favorite
classifier**
learned



"car"

Deep VISION

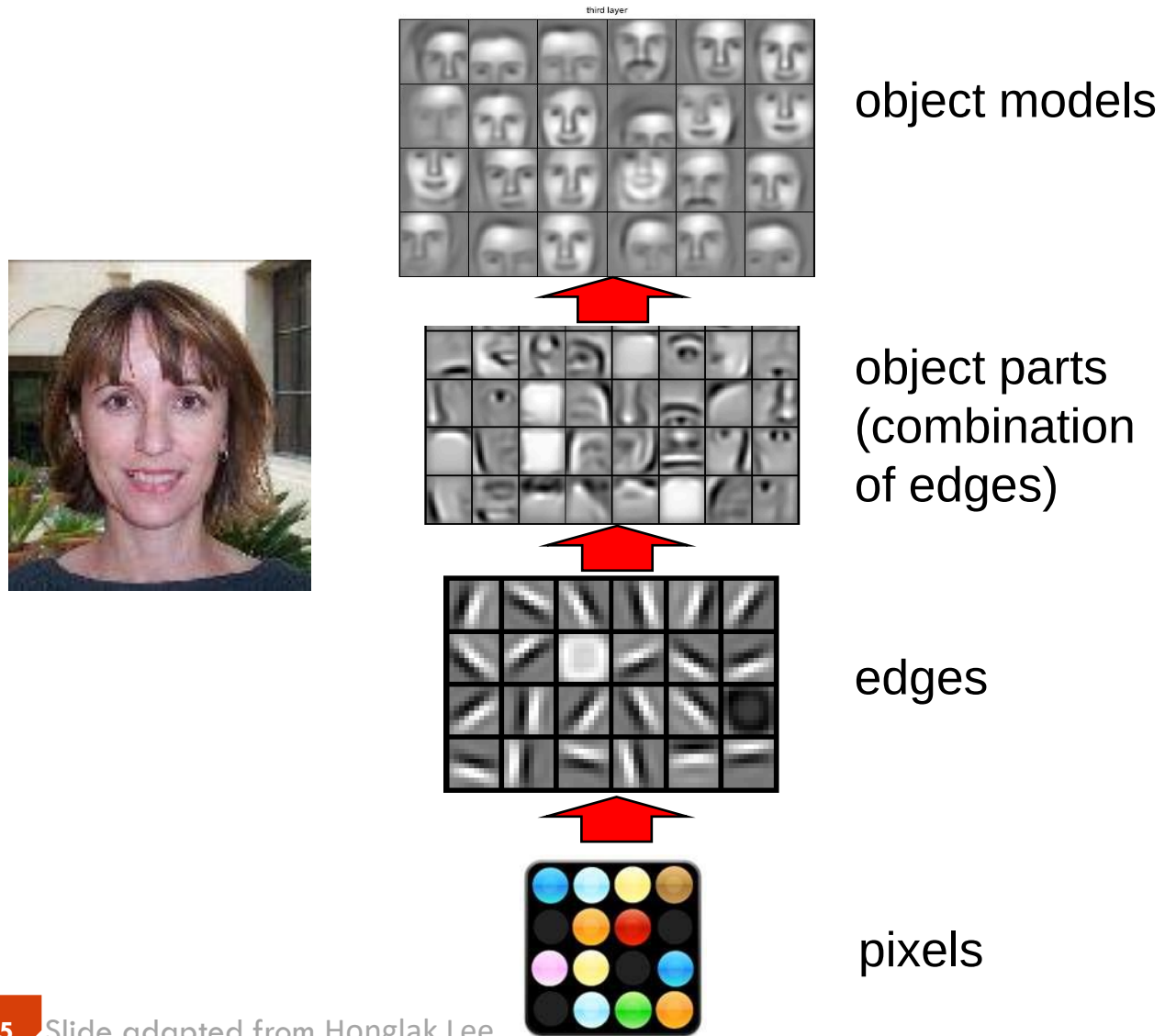


"car"

Sequence of differentiable non-linear
functions feeding to the next one



Deep Learning: Let's Learn the Representation!

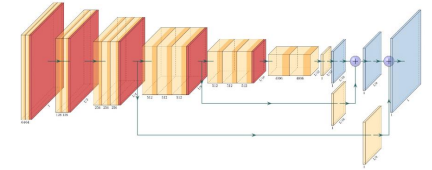




What is deep learning? Matching inductive bias to problem structure.



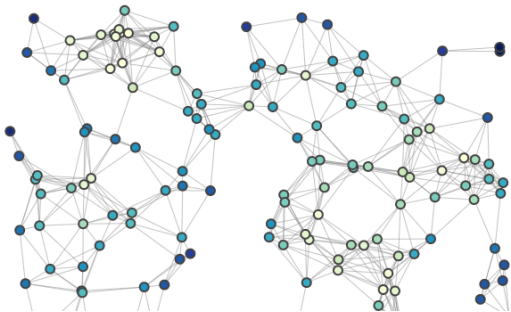
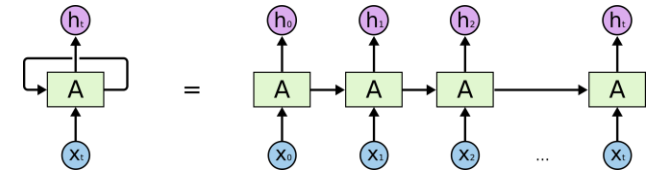
Translation Invariance
with Convolution



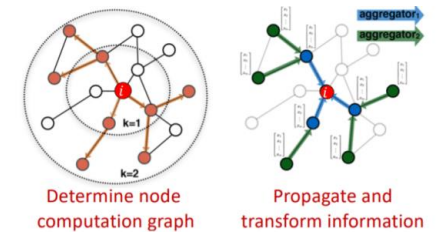
This burrito place
is yummy and fun!



Recurrence & Long-
Range Dependencies



Message Passing

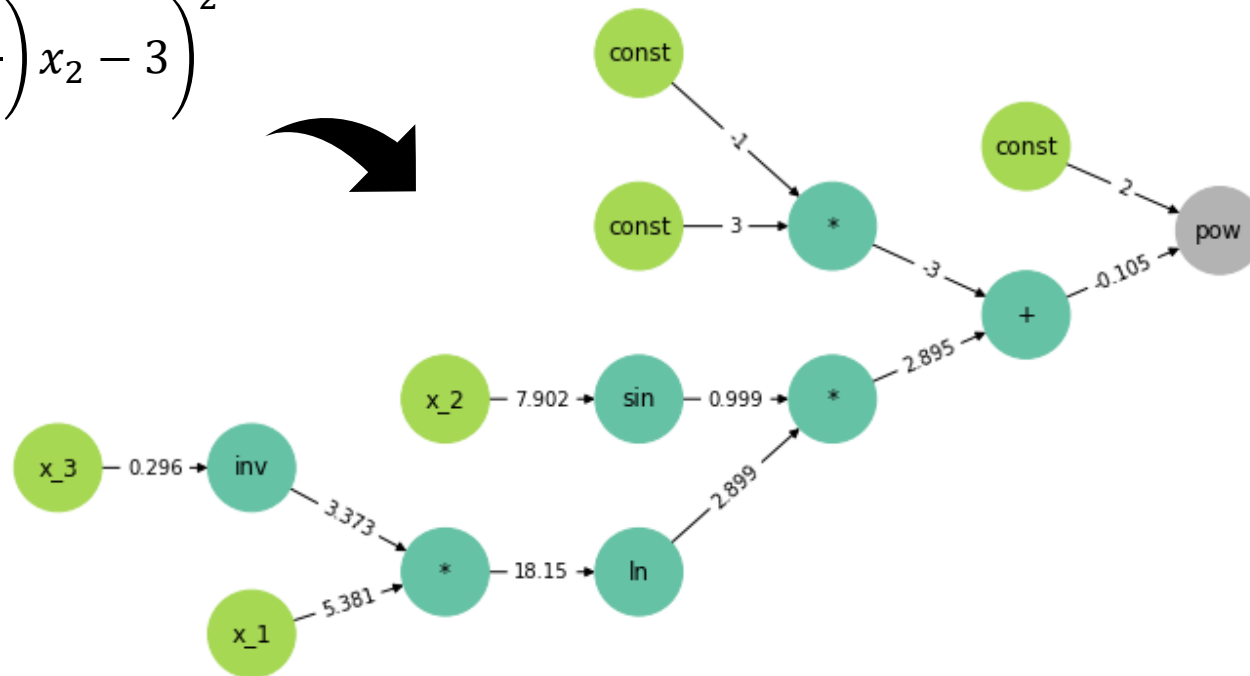




What is deep learning? Modularly composed differentiable functions.

Compose any sequence of differentiable operators to map from input to output - use gradient descent to optimize

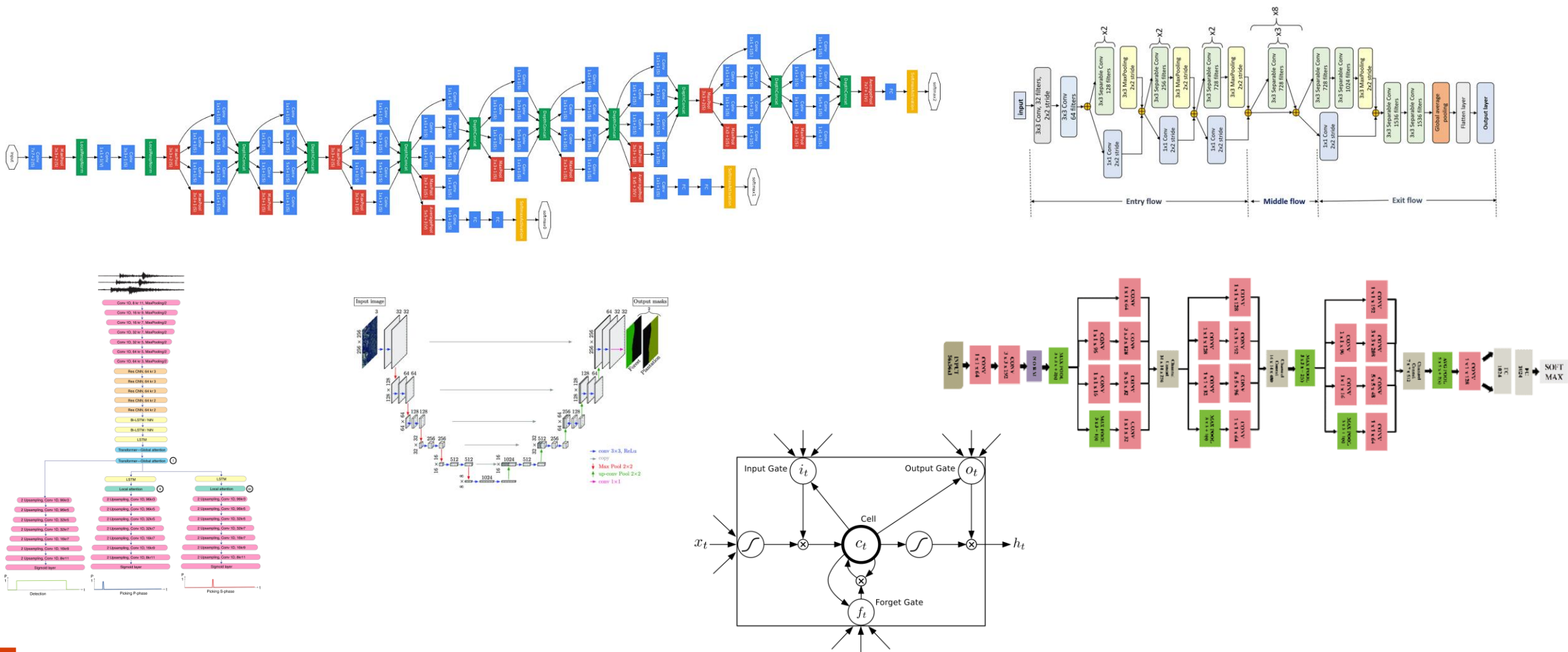
$$f(\mathbf{x}) = \left(\ln\left(\frac{x_1}{x_3}\right) x_2 - 3 \right)^2$$





What is deep learning? Modularly composed differentiable functions.

Compose any sequence of differentiable operators to map from input to output - use gradient descent to optimize

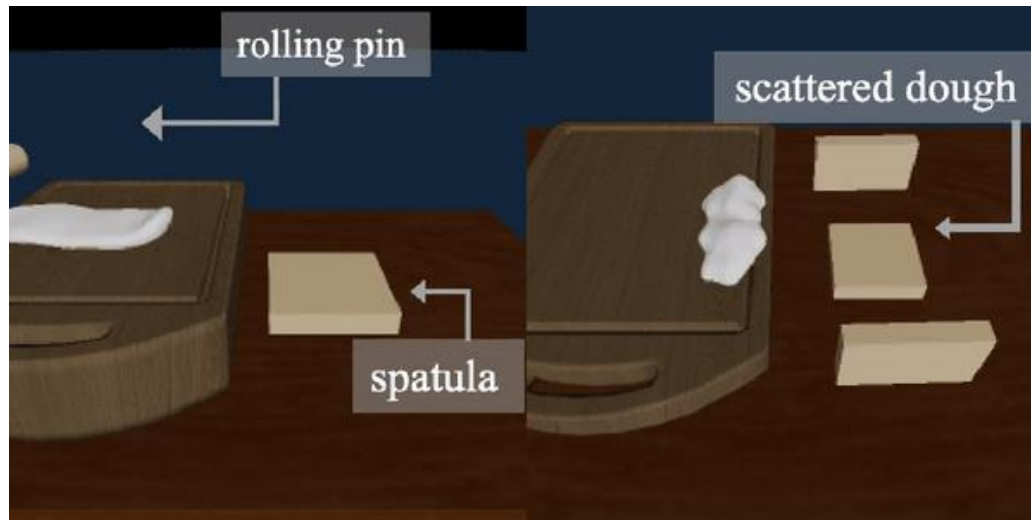
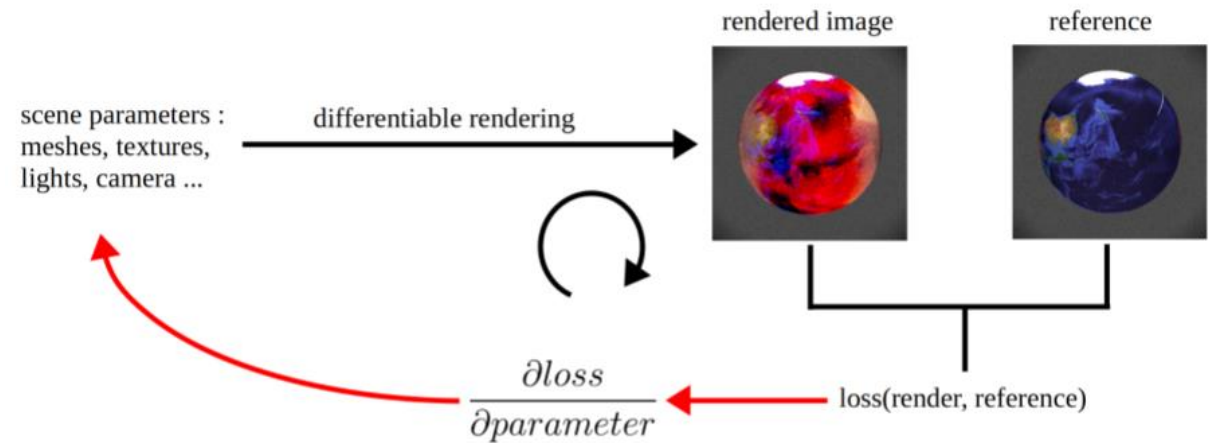




What is deep learning? Modularly composed differentiable functions.

Even starting to include non-learned but differentiable elements

Differentiable renderers to optimize physical properties



Differentiable physics engines to optimize control policies



What is deep learning? Underlying technologies.

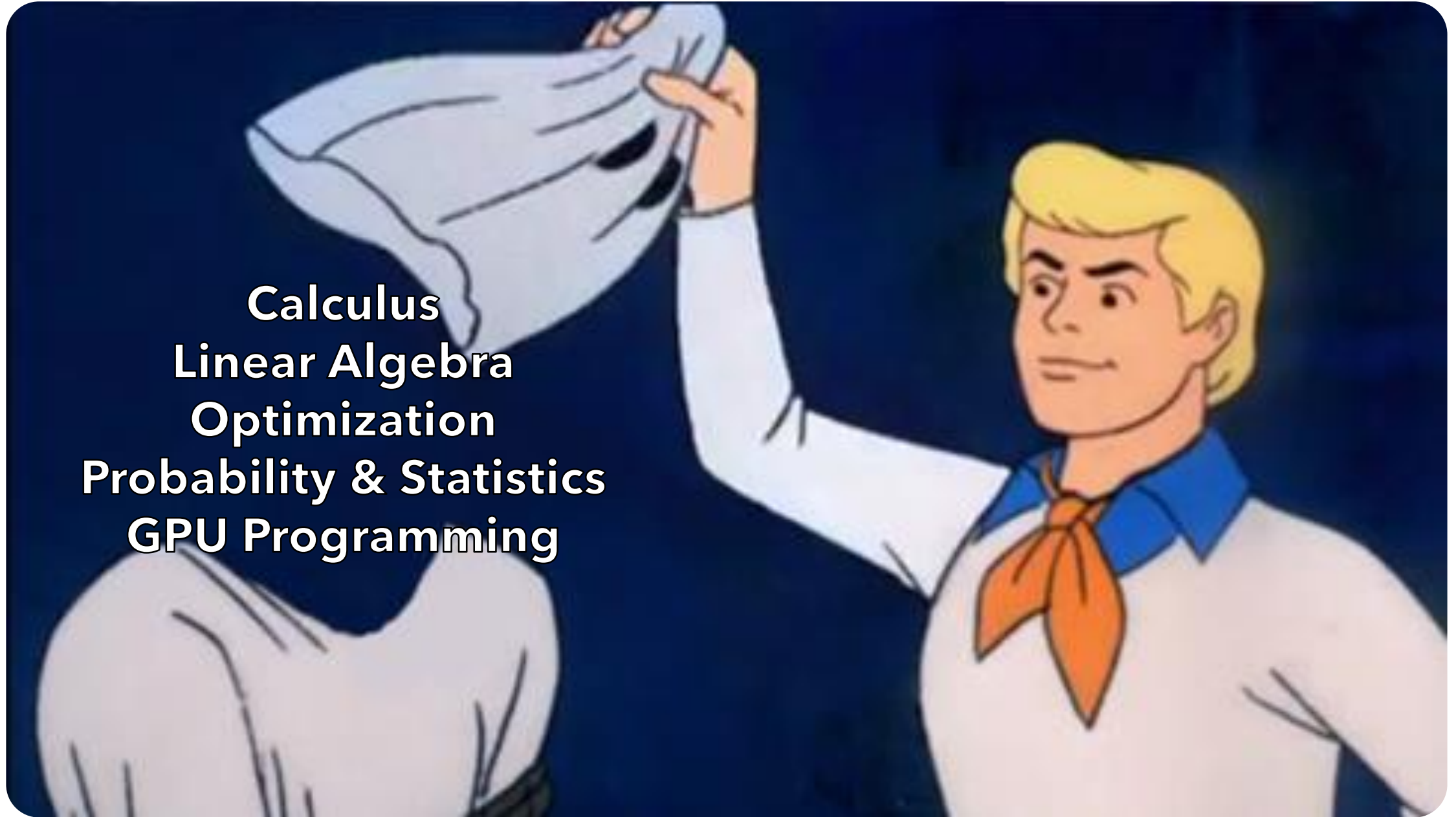
Deep Learning





What is deep learning? Underlying technologies.

Calculus
Linear Algebra
Optimization
Probability & Statistics
GPU Programming





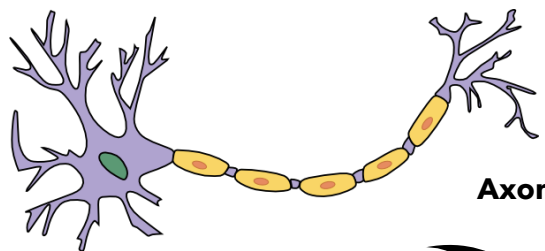
What is deep learning? History of booms and busts.



1957-1960s - The Perceptron

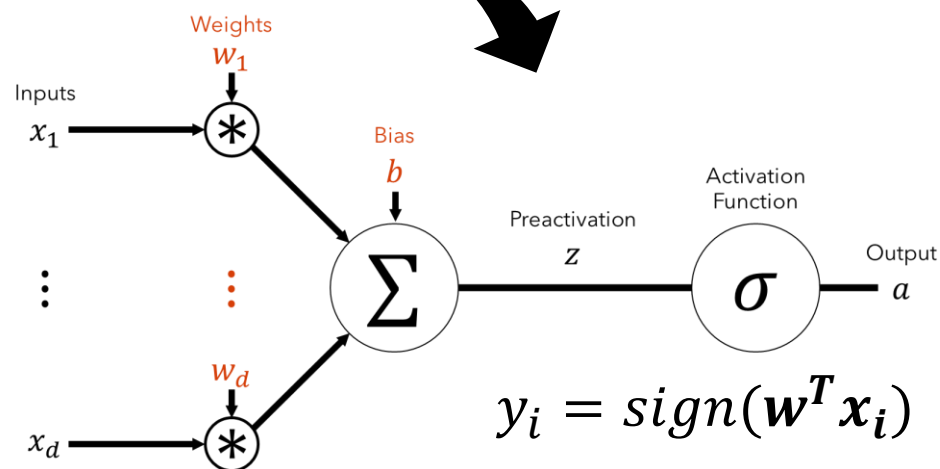
- Introduced in 1957 by Frank Rosenblatt

Dendrite



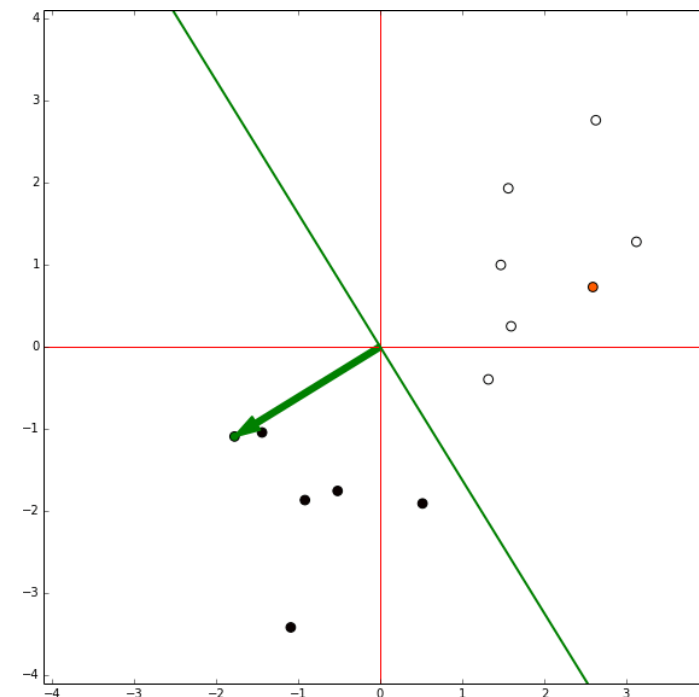
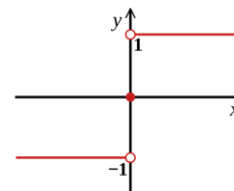
Axon

Inspired



Sign Function

$$\text{sign}(z) = \begin{cases} 1 & z \geq 0 \\ -1 & \text{else} \end{cases}$$



for each example $\mathbf{x}_i, y_i \in D$:

if $y_i \mathbf{w}^T \mathbf{x}_i < 0$: // if misclassification

$$\mathbf{w} = \mathbf{w} + \alpha y_i \mathbf{x}_i$$

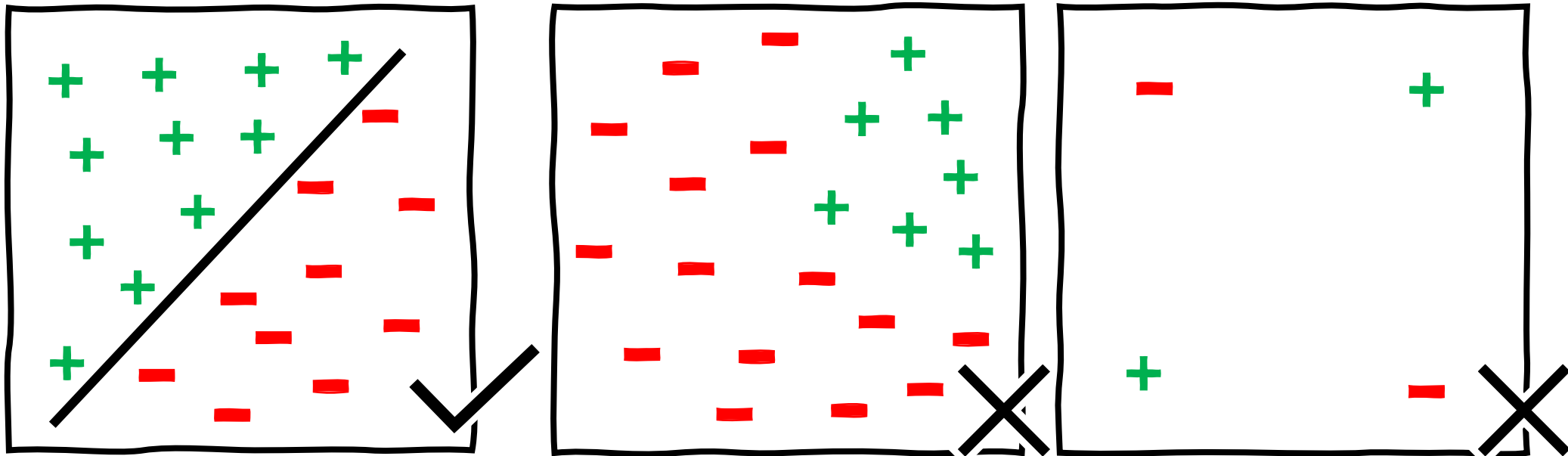


What is deep learning? History of booms and busts.



1970s - Hiatus

- Perceptrons (Minsky and Papert 1969)
 - Revealed the fundamental difficulty in linear perceptron models
 - Stopped mainstream research on this topic for more than 10 years



- Lots of expert systems followed...



What is deep learning? History of booms and busts.

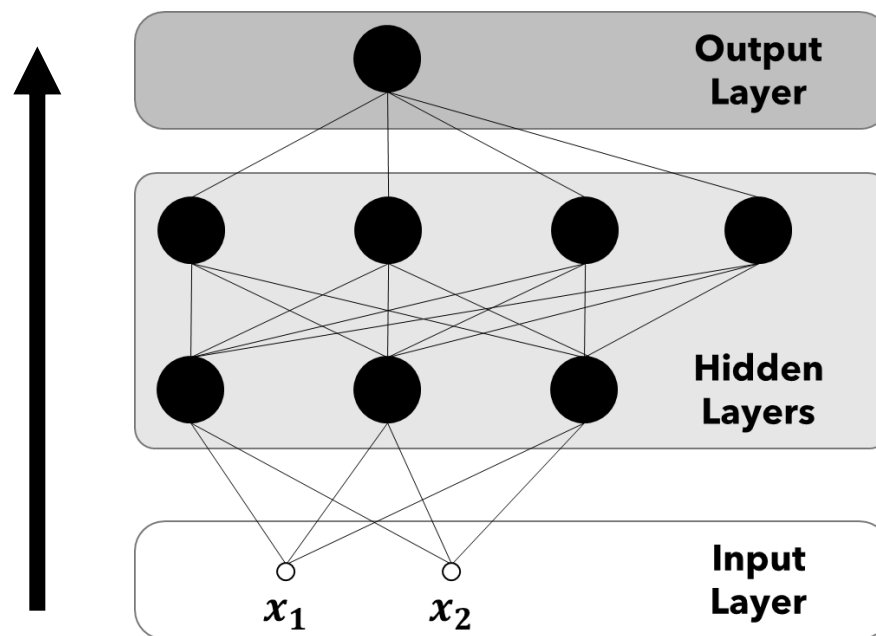


1980s - Nonlinear Neural Networks

- Rumelhart, Hinton, Williams 1986
 - Werbos 1974 did similar, but was mostly not recognized

Forward Pass:

Each neuron passes its output to ones ahead of it



Backward Pass:

Chain rule computes gradient from output to hidden layers



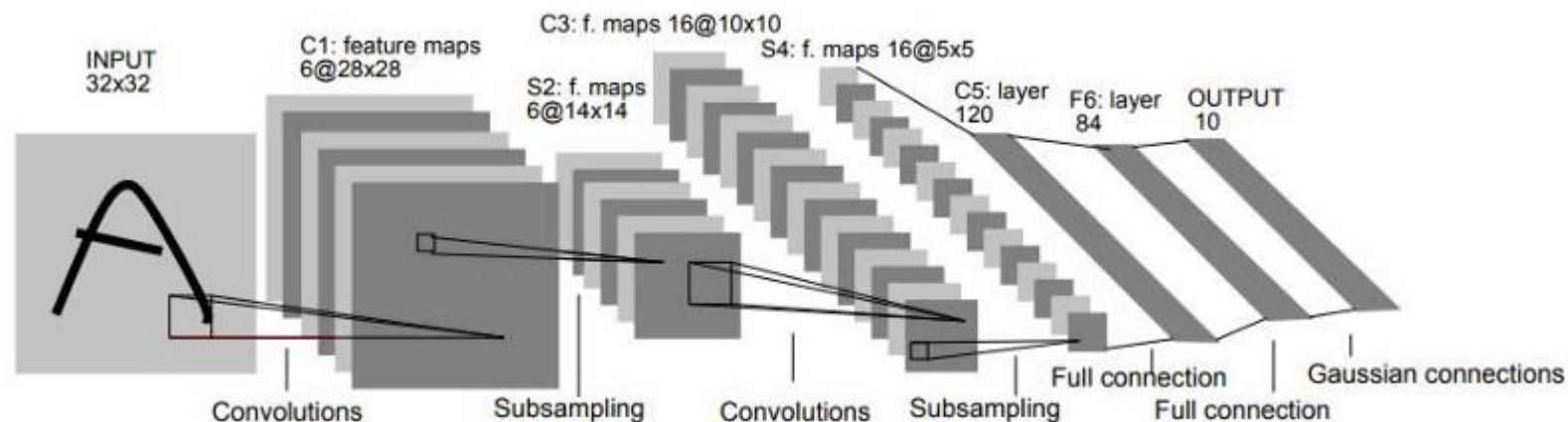


What is deep learning? History of booms and busts.



1990s - Early Successes

- Success in handwritten digits
- Boltzmann machines, LSTMs, networks of all sorts





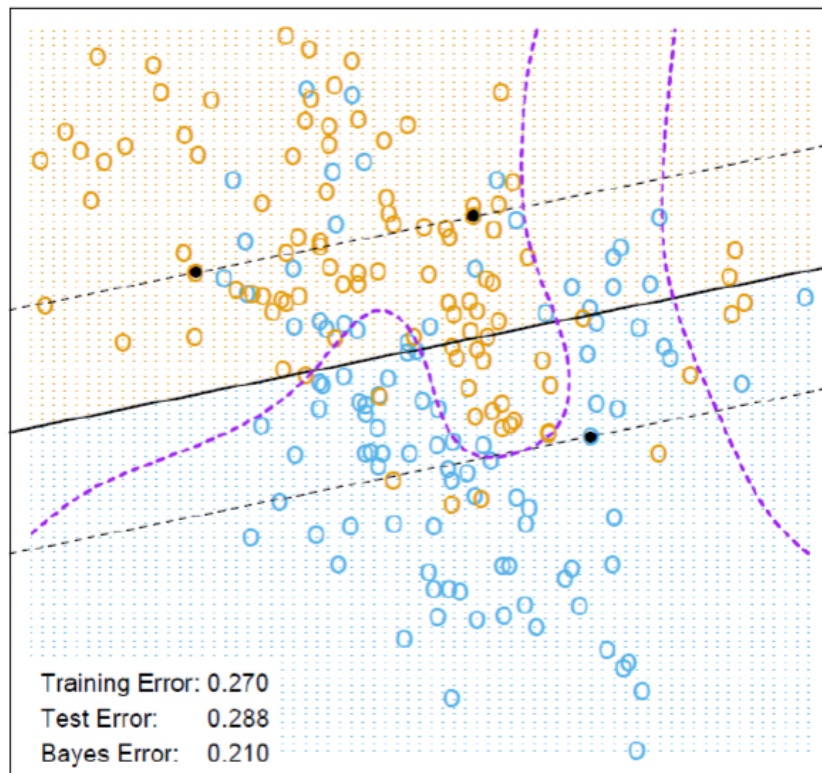
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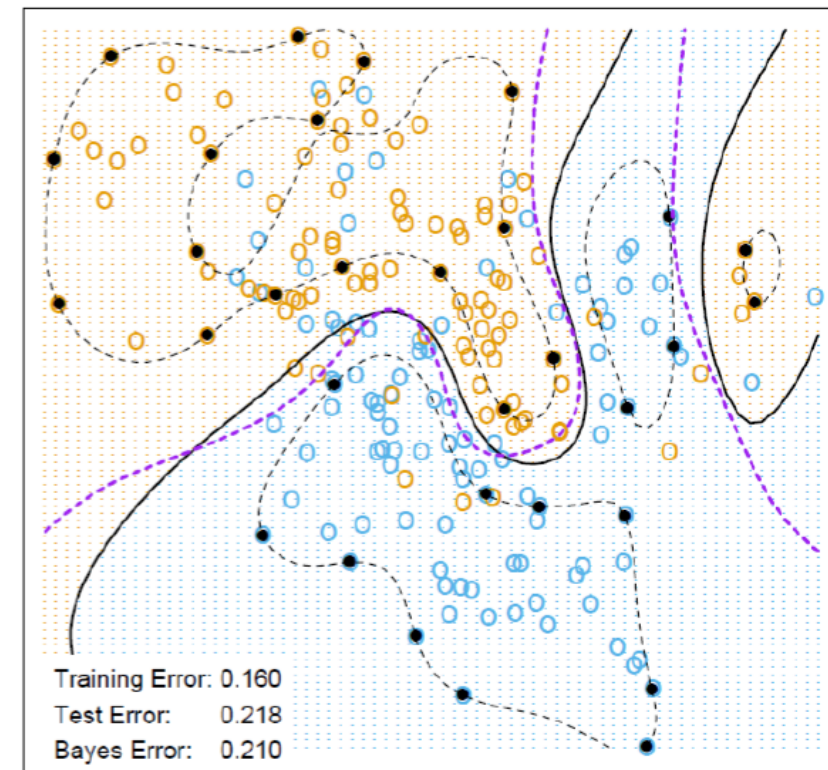
1992-2010s - Rise of Kernel Methods

- Cortes, Vapnik 1995. Vapnik 1995. Vapnik 1998.
- First paper forced to publish under "Support Vector Networks"

SVM: Linear



SVM - Radial Kernel in Feature Space





2010s - Don't Call It A Comeback

- Learn many, many layers simultaneously
 - Max-pooling (Weng, Ahuja, Huang 1992)
 - Stochastic gradient descent (Hinton 2002)
 - ReLU Nonlinearity (Nair and Hinton 2010, Krizhevsky, Sutskever, Hinton 2012)
 - Dropout (Hinton et al. 2012)
- WAY more labeled data available
 - Crowd-sourced annotation scaled data
- WAY more compute
 - GPU processing enables core linear algebra operations



Do I need this class?

When to use deep learning:

- Highly-structured input or output that can be well aligned with inductive biases in the model architecture
- Lots of data or convenient transfer domain
- Reasonable compute

Otherwise:

- Plenty of other theoretically universal approximators:
 - SVMs, Random Forests, Fourier Features



What are some problems with deep learning?

Non-convex! Non-convex! Non-convex!

- For depth ≥ 3 : most losses non-convex in parameters
→ theoretically, all bets are off in terms of optimality
- Leads to stochasticity
 - Different initializations → different local minima

Standard Response #1:

- “Yes, but all interesting learning problems are non-convex”

Standard Response #2:

- “Yes, but it works pretty often!”



What are some problems with deep learning?

Lack of interpretability

- Hard to track down what's failing

Standard Response #1:

- "We're working on it"
 - feature visualization, saliency maps, neuro-symbolic models

Standard Response #2:

- "Yes, but it works pretty often!"



What are some problems with deep learning?

Needs a lot of data

- Not all problems have that much data, networks overfit otherwise

Standard Response #1:

- “Transfer learning techniques are improving”
 - Train on a related domain with lots of data and then transfer
 - See “foundational models”

Standard Response #2:

- “Yes, but it works pretty often!”

Today's Learning Objectives

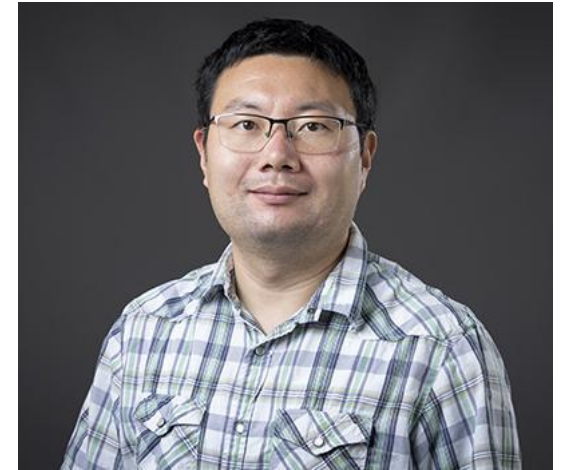
Be able to answer:

- ~~What is deep learning?~~
 - ~~Why and when should we care about it?~~
 - ~~What are some problems with deep learning?~~
-
- How is this class structured?
 - How will I be assessed / graded?





- **Instructor:** Fuxin Li
- **Class Period:** M/W 4-5:50PM Kidder 364
- **Office Hours:** One in-person and one on Zoom TBD. Participate in the whenisgood poll.
- **Class Webpage:** CANVAS
- **Questions/Discussions** – CANVAS?
- **Teaching Assistant:** (Office hours TBD)



Wesley Khademi



Nathan Funckes





- **Am I prepared for this course?**

- Are you comfortable with most of these:

- Machine learning foundations
- Manipulating probabilities and distributions
- Taking derivatives and reasoning about integrals
- Performing common matrix operations
- Programming
- Discrete Mathematics (Graphs)

[**Machine Learning**]

[**Probability**]

[**Calculus**]

[**Linear Algebra**]

[**Computer Science**]

[**Computer Science**]



Grading / Assessment:

- **Quizzes** (30%) - Quizzes given periodically **in-person** that cover course content.
- **Assignments** (25%) - Three assignments involving programming and mathematics. Programming will be in Python.
- **Project** (45%) - A self-designed project on deep learning performed in groups of 3. Three deliverables:
 - Initial Presentation (5%)
 - Final Presentation (5%)
 - Final Report (35%)



Book:

I. Goodfellow, A. Courville, Y. Bengio. Deep Learning. MIT Press 2016.

Electronic version: <http://www.deeplearningbook.org/>

More readings can be found at:

<http://deeplearning.net/reading-list/>

<http://colah.github.io/>

<http://karpathy.github.io/>

<https://www.coursera.org/course/neuralnets>

<https://towardsdatascience.com/>



A plethora of deep learning toolboxes:

Torch, pyTorch, TensorFlow, JAX, MXNet, CNTK, Lasagne, Keras, Neon, Caffe, etc.

A lot of convergence to pyTorch/JAX nowadays

Toolbox policy:

We stick to pyTorch for assignments

Final project: select the one you are most comfortable with



What will be covered

- Basic Neural Network Structure
- Training tricks (SGD, Momentum etc.)
- Deep Learning on Visual Data
- Deep Learning on Graphs and Point Clouds
- Unsupervised and Self-Supervised Learning
- We will deliberately **NOT** cover sequence models (and LLMs except for the GPT model)
 - Find that in the AI539: ST/NLP with Deep Learning
 - In the future we hope that to become Deep Learning II



Late Work Policy:

For assignments, each students have 7 "late days" that can be used during the term. Using a late day extends the deadline for an assignment by 24 hours. When submitting a late assignment, indicate in Canvas via a comment how many late days are being used. **No late quizzes or project deliverables will be accepted.**

Collaboration Policy:

Students are expected to work independently on quizzes and assignments. High-level discussion of topics is allowed, but any low-level details, sharing of code, or collaborative programming is prohibited. You **are** allowed to use copilot-type tools (including chatGPT) to aid your programming. However, you are responsible to make the code to run on its own.



- Groups of no more than 3 persons
- Jointly work on a significant project
 - Must use deep learning, must **train (or fine-tune)** a model
 - CANNOT be just testing an already-trained classifier
 - Try to solve a real problem
 - One can elect projects from paper readings
 - I will try to suggest some standard projects
 - Also check [www.Kaggle.com](https://www.kaggle.com)
 - New neural architectures/changes to current architectures are welcome (but not required)
- Grading - based on the project merit, execution and presentation



- 2 presentations for the final project
 - Initial design (at least 1 month before finals week)
 - Talk about what is your project about
 - What you plan to do
 - Re-grouping if several people are thinking about similar projects
 - Final presentation (finals week)
 - Need to identify who did what in the team
- 8 minutes per presentation
 - Present from your own computer



- Pelican cluster:
 - 4 nodes with 2 GTX 1080 Ti (11GB) each
 - Accessible by SSH at pelican.eecs.oregonstate.edu
 - Policy: 1 GPU per group otherwise risk your jobs be killed
- Compute-gpu4: 8 GTX 2080 Ti GPUs
- Colab: (<https://colab.research.google.com/>)
 - Free tier has NVIDIA T4 (slightly worse than 1080 Ti)
 - Pro tier (\$10 per month) has P100 (about similar with 1080 Ti)
- If you want to buy your own:
 - 3090 Tis and 4090 Tis are very good but expensive
 - 3060 Ti and 4060 Ti/4070s are good options
 - Always go for **more RAM**



AI535 - Rough Schedule

	Topics Covered	Quiz	Assignment	Project
Week 1	Introduction, Logistics, and Optimization Refresher		A1 Out.	
Week 2	Basic Feedforward Networks			
Week 3	Neural network Optimization (I) and Convolutional Networks (I)	Q1	A1 Due. A2 Out.	
Week 4	Convolutional Neural Nets (II)	Q2		
Week 5	Practicum on Tools for Projects + Generative Models (I)		A3 Out.	
Week 6	Generative Models (II)	Q3	A2 Due.	
Week 7	Project Proposals Presentations			Project Proposal Presentations
Week 8	Neural Network Regularization + Graph Neural Networks	Q4	A3 Due.	
Week 9	Point Cloud Networks + Theoretical Implications			
Week 10	Self-Supervision, Diffusion Models, Student-Elected Topics			
Week 11	--			Final Presentation and Report

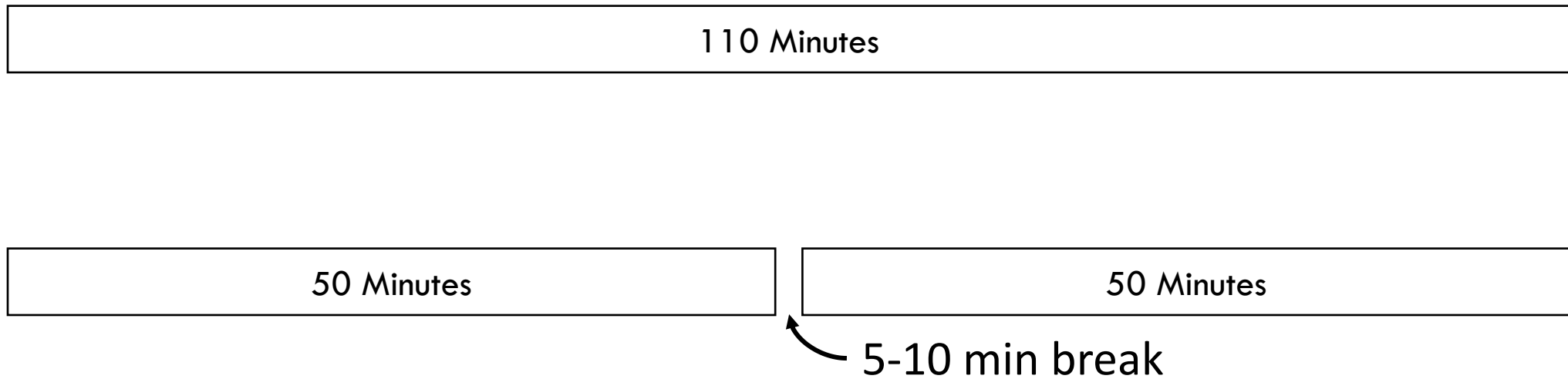


- **What this class is:**

- An introduction to the fundamentals of deep learning. You will do math to derive and **understand** deep learning algorithms. You will implement learning algorithms from scratch and apply them to provided datasets. You will work on a project.

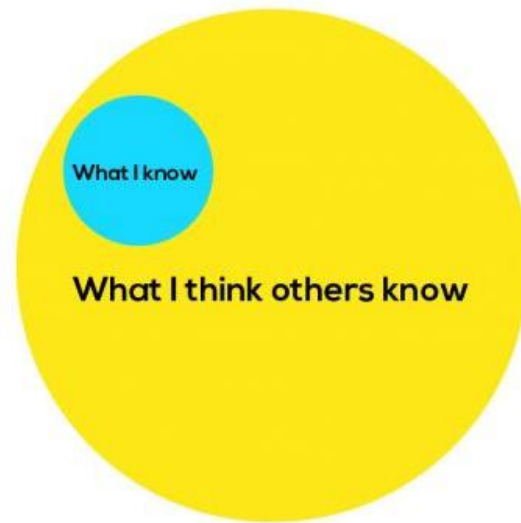
- **What this class isn't:**

- Not an in-depth exploration of the latest and greatest cutting edge
- Not an intro to machine learning class - you need to already know this bit.
- Not a practicum course on framework X, Y or Z - will provide resources, not focus.

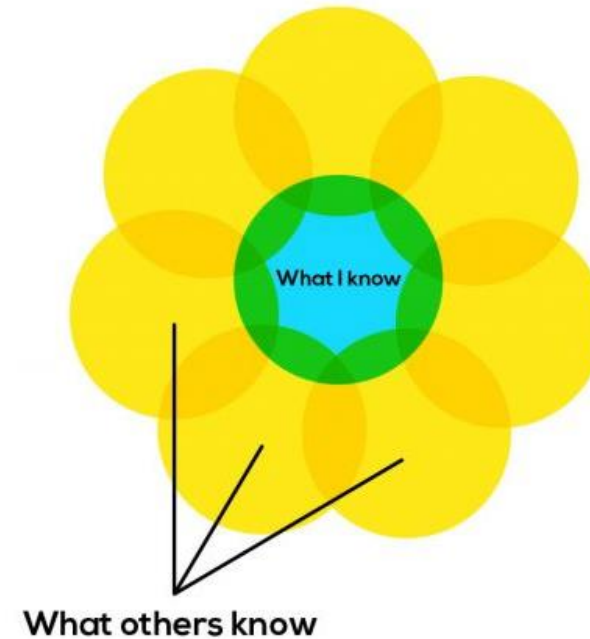


It is okay not to know something - just ask.

Imposter Syndrome



Reality



Today's Learning Objectives

Be able to answer:

- ~~What is deep learning?~~
- ~~Why and when should we care about it?~~
- ~~What are some problems with deep learning?~~

- ~~How is this class structured?~~
- ~~How will I be assessed / graded?~~





Next Time: We'll go more in-depth on optimization.